WORKSHEET 5.11

Behaviour of Implicitly Defined Curves

Unit 5: Analysing Functions

Key ideas

- Write $\dfrac{dy}{dx}$ as a single fraction. The tangent is **horizontal** where the numerator is zero (and the denominator is not).
- The tangent is **vertical** where the denominator is zero (and the numerator is not).
- A condition such as $y = 2x$ must be substituted back into the **original equation** to find actual points. Some conditions give no points.
- At a point with a horizontal tangent, the sign of $\dfrac{d^2y}{dx^2}$ decides between a maximum and a minimum.

Worked examples

Example 1 · Slope at a point

Find the slope of $x^2 + y^2 = 10$ at $(1, 3)$.

Solution. $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

$-\dfrac{1}{3}$.

Example 2 · Horizontal and vertical tangents

Find the points on the ellipse $4x^2 + y^2 = 16$ where the tangent is horizontal, and where it is vertical.

Solution. $8x + 2y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{4x}{y}$. Numerator zero: $x = 0$, $y = \pm 4$. Denominator zero: $y = 0$, $x = \pm 2$.

Horizontal at $(0, \pm 4)$; vertical at $(\pm 2, 0)$.

Example 3 · Maximum or minimum

For the same ellipse, find $\dfrac{d^2y}{dx^2}$ at $(0, 4)$ and interpret it.

Solution. Differentiate $8x + 2y\dfrac{dy}{dx} = 0$ again: $8 + 2\left(\dfrac{dy}{dx}\right)^2 + 2y\dfrac{d^2y}{dx^2} = 0$. At $(0, 4)$ with $\dfrac{dy}{dx} = 0$:

$8 + 8\dfrac{d^2y}{dx^2} = 0$.

$\dfrac{d^2y}{dx^2} = -1 < 0$**: the curve has a relative maximum at $(0, 4)$.**

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Practice

Questions 5 to 8 refer to the curve $y^3 - 3y = x^2$.

CORE

1. Find $\dfrac{dy}{dx}$ for $x^2 + y^2 = 10$.

2. Find the slope of that circle at the point $(3, -1)$.

3. At which points of the circle $x^2 + y^2 = 10$ is the tangent horizontal?

4. At which points of that circle is the tangent vertical?

EXAM LEVEL

5. Find $\dfrac{dy}{dx}$ in terms of x and y.

6. Find all points of the curve at which the tangent line is horizontal.

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7. Find all points of the curve at which the tangent line is vertical.

8. Verify that $\left(3\sqrt{2}, 3\right)$ lies on the curve, and find the slope there.

Question 9 · Free-response practice

Consider the curve $xy + y^2 = 8$.

(a) Find $\dfrac{dy}{dx}$ in terms of x and y.

(b) Write an equation of the line tangent to the curve at the point $(2, 2)$.

(c) Show that the curve has no horizontal tangent and no vertical tangent.

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Worked solutions

1. $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

2. $-\dfrac{3}{-1} = 3$.

3. Where $x = 0$: $\left(0, \sqrt{10}\right)$ and $\left(0, -\sqrt{10}\right)$.

4. Where $y = 0$: $\left(\sqrt{10}, 0\right)$ and $\left(-\sqrt{10}, 0\right)$.

5. $\left(3y^2 - 3\right)\dfrac{dy}{dx} = 2x$, so $\dfrac{dy}{dx} = \dfrac{2x}{3y^2 - 3}$.

6. $x = 0$ gives $y^3 - 3y = 0$, so $y = 0$ or $y = \pm\sqrt{3}$: the points $(0, 0)$, $\left(0, \sqrt{3}\right)$ and $\left(0, -\sqrt{3}\right)$.

7. $y = \pm 1$. $y = 1$ gives $x^2 = -2$ (no points); $y = -1$ gives $x^2 = 2$: the points $\left(\sqrt{2}, -1\right)$ and $\left(-\sqrt{2}, -1\right)$.

8. $27 - 9 = 18 = \left(3\sqrt{2}\right)^2$. The slope is $\dfrac{6\sqrt{2}}{24} = \dfrac{\sqrt{2}}{4}$.

9. (a) $y + x\dfrac{dy}{dx} + 2y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{y}{x + 2y}$.

   (b) The slope is $-\dfrac{2}{6} = -\dfrac{1}{3}$: $y - 2 = -\dfrac{1}{3}(x - 2)$.

   (c) Horizontal needs $y = 0$, but then $0 = 8$: no such point. Vertical needs $x = -2y$, which gives $-2y^2 + y^2 = 8$, that is $y^2 = -8$: no such point.

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